

论文信息

订单号:	226082800292972072
论文编号:	ME2UYEMHWY
论文名称:	GBGCL: Granular Ball Graph Contrastive Learning with Ball-Aware Topology Encoding
关键词:	graph contrastive learning, representation scattering, granular ball computing, self-supervised learning

评审意见

评审意见: 返修

评审时间: 2026-08-28

1、Completeness of Technical Content

- ☐ Experiment is complete with excellent data analysis and conclusion
- ☒ Experiment is complete with good data analysis and conclusion
- ☐ Experiment is complete with ordinary data analysis and conclusion
- ☐ Experiment is incomplete with insufficient data, ordinary analysis and conclusion
- ☐ Experiment is poor, with insufficient data and no analysis or conclusion

2、Scientific Significance

- ☐ The research is of excellent significance and makes great contributions to the development of the discipline
- ☐ The research is of good significance and makes general contributions to the development of the discipline
- ☒ The research is of ordinary significance and makes a little contributions to the development of the discipline
- ☐ The research is of poor significance and makes little contributions to the development of the discipline
- ☐ The research is of no value for research

3、Scientific Rigor of Data

- ☒ Data and analysis are good to support the research conclusion
- ☐ Data and analysis are ordinary to support the research conclusion
- ☐ Data and analysis cannot support the research conclusion

4、References

- ☒ References are good to support the theoretical research
- ☐ References are ordinary to support the theoretical research
- ☐ References cannot support the theoretical research

5、Conclusion

This paper addresses the issue that SGRL neglects mesoscopic communities, and proposes GBGCL. The authors construct quality-guided granules and diffuse them in the graph, and inject the mesoscopic topology into the dual encoder using BTCM. Experiments show that it outperforms SGRL slightly on four benchmarks, with the highest improvement of 0.09 percentage points. The ablation study indicates that BTCM is the key, but the gain is limited and the innovation is small. However, there are some issues that need to be addressed before the paper can be published. Please see the comments section for details.

6、Comment

The paper lists the convex write-back loss, the spherical-level dispersion loss, and the spherical-level InfoNCE loss as components of the particle-sphere diffusion module. However, the node representations generated by the convex write-back process are not included in the objective function, nor are they used in the online and prediction heads when evaluating. 2. The paper only provides the partition criteria for each dataset, similarity measurement, and write-back weights, but does not report the reconstruction cycle of granules, the number of steps in multi-step diffusion, the number of granules, and the homogeneity or purity threshold used for recursive splitting. 3. The mechanism diagnosis only provides the cosine similarity between node embeddings and write-back vectors, as well as the difference ratio. It does not report the changes of dispersion loss, alignment loss, or topological constraint terms during training, nor does it provide gradient analysis. Therefore, the claim that global dispersion and granule cohesion are reconciled without direct support from the training dynamics level is not convincing. 4. Some sentences are abnormally truncated at the end of paragraphs or chapters, lacking subsequent predicates, objects, or modifiers, resulting in incomplete sentence meanings. 5. There are unclear or incomplete reference pronouns in subordinate clauses, as well as errors in inserted phrases interrupting the subject-predicate combination of the main clause, causing ambiguity in the logical subject and missing objects in some long sentences. 6. Please follow the paper template on the official website of the conference to typeset.

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